

DOMAIN DATA & KNOWLEDGE MICROCOSM BLUEPRINT

HealthTech — Medical Claims & Payer Auditing

DSBA 6171 — Assignment 1

Structured Data • Knowledge Corpus • Governance Metadata • AI-Ready Decision Support

Table of Contents

1. Business Scenario & Decision Question.....	3
2. Structured Data Architecture.....	4
3. Knowledge Corpus & Metadata.....	6
4. End-to-End Domain Microcosm Diagram.....	8
5. Business Risk Matrix & Quality Issues.....	8
6. Structured-to-Knowledge Linkages.....	9
7. Assignment Goals.....	9
8. Key Design Principles.....	10
Conclusion.....	10

1. Business Scenario & Decision Question

This microcosm represents the operational environment of medical claims review and payer auditing within a mid-sized health insurance organization. Payers receive thousands of claims daily from hospitals, clinics, and individual providers. Each claim must be evaluated against member eligibility, plan benefits, procedure codes, medical necessity criteria, prior authorization requirements, and jurisdiction-specific regulations.

Stakeholders

- Claims processors
- Clinical review nurses
- Provider relations teams
- Compliance and regulatory teams
- Fraud, Waste & Abuse (FWA) investigators
- Members and providers affected by claim decisions

Primary Business Problem

Claim decisions are often inconsistent due to:

- Complex and frequently changing coverage rules
- Varying interpretations of policy language
- Lack of point-in-time document retrieval
- Inconsistent application of medical necessity criteria
- Jurisdiction-specific rules that override national policies

The microcosm provides a controlled environment to explore how structured data, knowledge retrieval, and governance can support more accurate, consistent, and explainable claim decisions.

Primary Decision Question

DECISION QUESTION

Given the member's plan coverage, the submitted procedure code, and current payer policies, should this claim be approved, denied, or escalated for clinical review?

Supporting Questions

1. Is the member eligible for coverage on the date of service?
2. Is the submitted procedure covered under the member's plan?
3. Does the claim satisfy medical necessity criteria?
4. Does the procedure require prior authorization?
5. Are the procedure and diagnosis codes valid and appropriately related?
6. Are there jurisdiction-specific rules affecting this claim?
7. Does the claim contain fraud, waste, or abuse indicators?

8. Should the claim be escalated for clinical review?

2. Structured Data Architecture

The microcosm includes six structured datasets stored in `data/structured/raw/`. These datasets model members, providers, plans, procedures, claims, and claim line items.

Datasets

File	Description
patient_accounts.csv	Member enrollment and plan information
providers.csv	Provider/facility directory
insurance_plans.csv	Plan benefits and coverage attributes
procedure_catalog.csv	CPT/HCPCS procedure reference
claims_ledger.csv	Claim-level header information
claim_line_items.csv	Line-level detail for each claim

Keys & Relationships

- Members → Plans (`plan_id`)
- Members → Providers (`primary_provider_id`)
- Claims → Members (`member_id`)
- Claims → Providers (`provider_id`)
- Claims → Plans (`plan_id`)
- Line Items → Claims (`claim_id`)
- Line Items → Procedures (`procedure_code`)

Time Elements

- Member coverage dates
- Plan effective/termination dates
- Claim received date
- Service start/end dates
- Line-level service dates

These support point-in-time analysis and historical retrieval.

ERD Reference

The ERD diagram (`architecture/domain_microcosm_erd.png`) illustrates all tables, keys, and cardinalities.

Intended Analytical Use

The structured layer supports:

- Eligibility checks
- Coverage determination
- Medical necessity evaluation
- Provider network validation
- FWA detection
- Point-in-time policy retrieval
- Downstream decision support and explainability

3. Knowledge Corpus & Metadata

The knowledge corpus contains 8 documents stored in data/documents/source/, including:

- Plan benefits booklets
- Clinical criteria guidelines
- Prior authorization matrices
- State-mandated benefits addenda
- Regulatory guidelines
- Medical necessity procedures
- Synthetic internal memos

Document Inventory

Doc ID	Title	Type	Authority Status	Effective Date	Jurisdiction
HLT-POL-001	Gold PPO Plan Benefits Booklet (2025)	Benefits Summary	Current	2025-01-01	National
HLT-POL-002	Silver HMO Plan Benefits Booklet (2024)	Benefits Summary	Superseded	2024-01-01	CA; NY; TX
HLT-REG-101	Medicare Secondary Payer (MSP) Guidelines	Regulatory Guideline	Current	2025-01-01	US
HLT-PROC-005	Medical Necessity Review Procedure	Clinical Procedure	Approved	2025-06-01	National
HLT-PROD-01	Prior Authorization Requirements Matrix	Guideline	Draft	2026-07-01	National
HLT-FAQ-003	Claims Appeals Process FAQ	FAQ	Current	2025-11-15	National
HLT-CLIN-009	Clinical Criteria for Common Procedures	Clinical Guideline	Current	2025-09-01	National
HLT-STATE-CA	California State Mandated Benefits Addendum	Memo	Current	2026-01-01	CA

Format Diversity

- Multi-page PDFs (benefits booklets, clinical guidelines)
- Markdown documents (prior authorization matrix)
- Text files (FAQs, regulatory guidance)
- Documents with substantial tables and hierarchical structures

Version & Authority Diversity

- Current vs. Superseded: Current plan booklet (HLT-POL-001) vs. superseded 2024 version (HLT-POL-002)

- Draft vs. Approved: Draft prior authorization policy (HLT-PROD-01) vs. approved clinical procedure (HLT-PROC-005)
- Regulatory vs. Internal: Modeled-after CMS/Medicare guidelines alongside synthetic internal policies

Metadata Schema

Each document includes a metadata record in metadata/document_metadata.csv:

Field	Purpose
doc_id	Unique document identifier
title	Document title
doc_type	Classification (Benefits Summary, Regulatory Guideline, etc.)
source	Origin (Synthetic Internal, Modeled after CMS, etc.)
owner	Responsible team (Product Management, Compliance, Clinical Operations)
effective_date	Date document becomes/became effective
version	Version number or release identifier
jurisdiction	Applicable geographic scope (National, US, CA; NY; TX, etc.)
authority_status	Governance state (Current, Superseded, Draft, Approved)
confidentiality	Access level (Public, Internal)
access_group	Authorized user groups (Claims Adjudicators, Clinical Reviewers, All)

This metadata supports:

- Governance and freshness checks
- Retrieval filtering by authority and jurisdiction
- Access control and confidentiality enforcement
- Version and currency validation
- Explainability (why this document was selected)

4. End-to-End Domain Microcosm Diagram

The microcosm diagram (architecture/domain_microcosm_erd.png) illustrates:

- Structured datasets: Members, providers, plans, claims, procedures
- Knowledge corpus: Policies, guidelines, addenda, regulatory materials
- Metadata layer: Document attributes, authority, versioning, governance
- Linkages: Structured signals that route to appropriate policy documents
- Quality issues: Controlled data and knowledge problems that demonstrate governance and validation challenges

This architecture shows how structured claim events are evaluated against the correct policy documents at the correct point in time.

5. Business Risk Matrix & Quality Issues

The microcosm includes three controlled quality issues documented in metadata/quality_issues_log.csv. These demonstrate how data and knowledge problems can affect downstream AI outputs and business decisions.

Quality Issues

Issue ID	Category	Layer	Description	AI Impact	Business Impact	Future Control
QI-001	Document Authority Status	Knowledge Corpus / Governance	Draft, unlinked document (HLT-PROD-01)	AI may retrieve draft rules as authoritative	Incorrect claim decisions; audit exposure	Enforce authority_status = Current filter
QI-002	Document Version / Currency	Knowledge Corpus / Document Lifecycle	Superseded plan booklet (HLT-POL-002)	AI may apply outdated coverage rules	Wrong cost-share; regulatory risk	Point-in-time retrieval logic
QI-003	Duplicate Claim Event	Structured Data / Ingestion	Near-duplicate claim (same member, provider, plan, service dates; different claim_id)	Double-counting; duplicate evidence	Overpayment; distorted FWA signals	Idempotency keys + deduplication

Pedagogical Purpose

These controlled issues teach:

- Governance gates: How metadata constraints prevent silent propagation of unreliable documents
- Point-in-time retrieval: How service dates must match document effective periods
- Data validation: How idempotency and deduplication protect against ingestion errors

6. Structured-to-Knowledge Linkages

The microcosm includes a structured linkage registry in metadata/structured_to_knowledge_linkages.csv that maps claim signals to policy documents. Examples:

- Claims for Gold PPO members → HLT-POL-001 (Gold Plan Benefits, Current)
- Claims for Silver HMO members with service dates in 2024 → HLT-POL-002 (Silver Plan Benefits 2024, point-in-time match)
- Any claim with secondary insurance → HLT-REG-101 (MSP Guidelines)
- Claims in California → HLT-STATE-CA (CA Mandated Benefits Addendum)
- Any claim requiring medical necessity evaluation → HLT-PROC-005 + HLT-CLIN-009

This linkage layer is central to building AI-assisted decision support that routes claims to the correct policy knowledge.

7. Assignment Goals

Assignment 1 — Domain Microcosm Blueprint

- Define a realistic healthcare domain problem
- Architect a small but complete dataset ecosystem
- Establish structured data foundations
- Identify governance and quality challenges
- Create a foundation for AI-assisted decision support

Downstream Assignments

Assignment	Focus
Assignment 2	Ingest, validate, and load structured data into a database
Assignment 3	Build knowledge corpus ingestion and retrieval pipelines
Assignment 4	Implement claim decision logic using structured data and retrieved knowledge, with explainability and governance layers

8. Key Design Principles

1. Realism: Modeled on actual healthcare claims workflows, regulations, and challenges
2. Completeness: Includes structured data, knowledge corpus, and metadata — all three layers needed for AI
3. Governance: Metadata constraints (authority_status, effective_date, access_group) demonstrate why data governance matters
4. Tractability: Small enough to manage in a course project; large enough to explore real challenges
5. Intentional Issues: Controlled quality problems teach validation, retrieval, and governance without disrupting core learning
6. Scalability: Design patterns are extensible to larger healthcare domains and other regulated industries

Conclusion

SUMMARY

This Domain Data & Knowledge Microcosm provides a complete, realistic, and pedagogically purposeful foundation for exploring data engineering, knowledge retrieval, governance, and AI-assisted decision support in a healthcare context. The combination of structured claims data, policy knowledge, metadata governance, and intentional quality challenges creates an environment where learners can practice end-to-end data engineering and AI practices in a meaningful domain.